

TOI-1516b

R-2022

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1516_251002 · created 2026-08-08T00:15:27+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460950.863 +/- 0.025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.029
Transit depth (Rp/Rs)^2	1.03 +/- 0.6 [%]
Orbital Inclination (inc)	85.3 +/- 1.43
Airmass coefficient 1 (a1)	1.1846 +/- 0.0049
Airmass coefficient 2 (a2)	-0.13 +/- 0.043
Scatter in the residuals of the lightcurve fit is	0.41 %
Best Comparison Star	None
Optimal Aperture	3.64
Optimal Annulus	7.2
Transit Duration (day)	0.063 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.102 ± 0.029 vs 0.1224 (deviation 0.7σ)
Duration fit vs published	0.063 d vs 0.1217 d (deviation 2.55σ)
Mid-time O–C	-7.8 min
Depth SNR	3.5
Residual scatter	0.41 %

Light curve

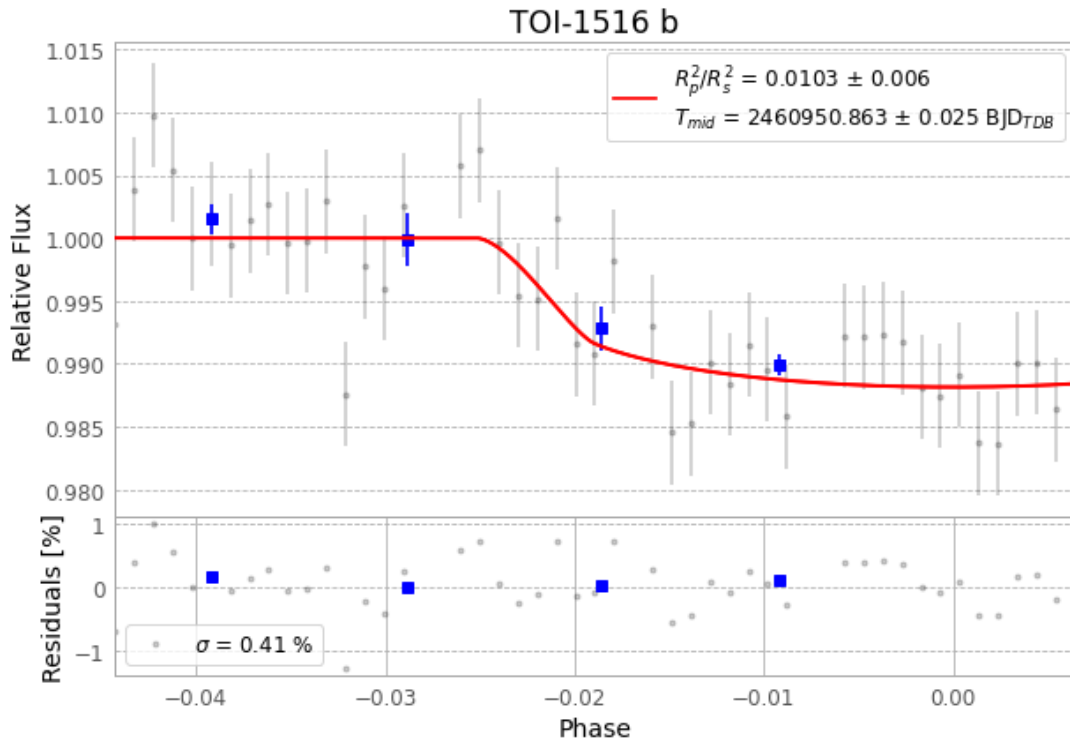


Figure 1 — FinalLightCurve_TOI-1516 b_2025-10-01.png (SHA-256 7f5b3e9e0907597a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 337], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f41c1a3894db4a70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

47 FITS frames (31 MB), TOI1516251002062715.FITS → TOI1516251002085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1516-251002.log (20a08068ff9f...), FinalLightCurve_TOI-1516 b_2025-10-01.png (7f5b3e9e0907...), FinalLightCurve_TOI-1516 b_2025-10-01.csv (601626f569da...)

Compute resources

Wall time 325.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:15Z.